

Week 1 Weekend Assignment

"Network Detective Challenge"

Due: End of Week 1 (before Day 5 test) | Time estimate: 3-4 hours across Saturday + Sunday

What you need: Your own laptop/PC, Wireshark (free), internet connection, this document

The Big Picture

Over the past 4 days you learned:

- Day 1: What cybersecurity is — CIA Triad, AAA, threats vs vulnerabilities vs risks
- Day 2: Networking basics — OSI model, TCP/IP, IP addressing, ports, protocols
- Day 3: Common protocols — HTTP/HTTPS, FTP/SFTP, SSH, SMTP, DNS, ICMP + subnetting
- Day 4: Linux & Windows security — permissions, users, hardening, event logs

Now it is time to combine all of that into one real-world mini-project.

This assignment has 3 parts. Do all 3. They build on each other.

Part 1 — Capture & Analyze HTTP Traffic (Day 2 + Day 3 skills)

What to do:

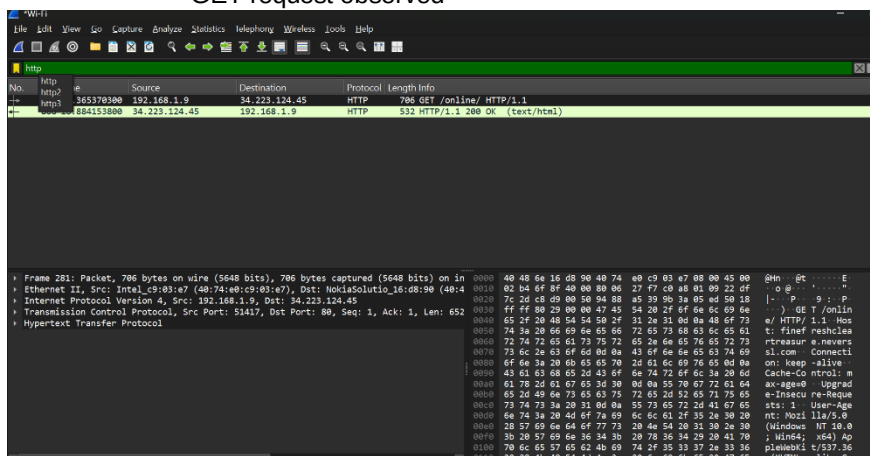
1. Install Wireshark (if not already): <https://www.wireshark.org/download.html>
2. Open Wireshark and start capturing on your active network interface (Wi-Fi or Ethernet).
3. While capturing, open a browser and visit 3 different websites:
 - One that uses HTTP (try <http://neverssl.com> — it deliberately uses HTTP)
 - One that uses HTTPS (try <https://www.google.com>)
 - One of your choice (any site you normally use)
4. Stop the capture after about 60 seconds of browsing.
5. Apply the display filter `http` in Wireshark — this shows only HTTP traffic.
6. Also try the filter `dns` to see DNS queries.
7. Also try the filter `tcp.port == 443` to see HTTPS traffic.

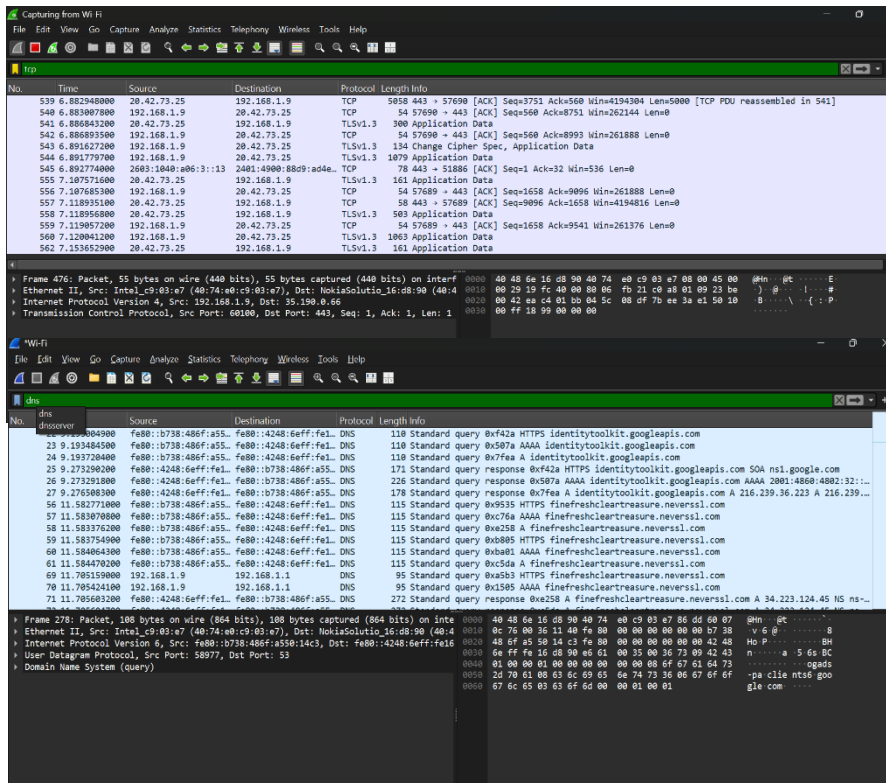
What to write about (answer in your own words):

For each of the 3 websites, answer:

HTTP Capture

- Protocol: HTTP
- Port: 80
- Traffic visible in plain text
- GET request observed





DNS Capture

- Protocol: DNS
- Port: 53
- Domain lookups observed
- Example domains:
 - identitytoolkit.googleapis.com
 - finefreshcleartreasure.neverssl.com

HTTPS Capture

- Protocol: HTTPS (TLS 1.3)
- Port: 443
- Traffic encrypted
- Application data not readable

Question	Your answer
What website did you visit?	
HTTP or HTTPS? How can you tell from the capture?	
What IP address did your computer connect to?	
What was the destination port? (80 or 443?)	
For HTTP sites: what information is visible in the packet?	
For HTTPS sites: what information is visible vs hidden?	
What DNS queries did you see before the connection?	

Key concepts to connect:

- CIA Triad: When you visited the HTTP site, which part of CIA was violated? (Hint: no encryption = no

Confidentiality)

- **Confidentiality:** HTTPS encryption protects sensitive information.
 - **Integrity:** TLS helps prevent modification of data in transit.
 - **Availability:** Web services remain accessible through secure connections.
-
- AAA: If the HTTP site had a login page, which AAA function would be exposed? (Hint: Authentication credentials

in clear text)

If the HTTP site had a login page, the Authentication function of AAA would be exposed. Since HTTP does not encrypt data, usernames and passwords travel in plain text. An attacker capturing packets could view the login credentials and impersonate the user. This breaks Authentication because the attacker can pretend to be a legitimate user.

• Subnetting: What is the IP range of your home network? (Check with ipconfig or ifconfig) What subnet mask are you using?

My network IP address is 10.61.36.71 and my subnet mask is 255.255.255.0. Therefore, my network is 10.61.36.0/24. The usable host range is 10.61.36.1 to 10.61.36.254.

Where to get help / sources:

- Wireshark official docs: <https://www.wireshark.org/docs/>
- Wireshark 101 tutorial: <https://www.youtube.com/watch?v=TkCSr30UojM>
- How to filter HTTP traffic: In Wireshark, type http in the display filter bar

Part 2 — Subnetting Practice (Day 3 skills)

Using what you learned about subnetting in Day 3, solve these problems. Show your working.

Problem 1

Your home router gives you the network 192.168.1.0/24.

- How many total host addresses are available in this network?

Total host addresses available, $2^{(32-24)} - 2 = 2^8 - 2 = 254$ usable hosts

- If you split it into 4 equal subnets, what is the new subnet mask in CIDR notation?

New subnet mask after splitting into 4 equal subnets:

Borrow 2 bits from host portion.

New subnet mask = /26

- What are the network addresses of all 4 subnets?

Subnet 1: 192.168.1.0/26

Subnet 2: 192.168.1.64/26

Subnet 3: 192.168.1.128/26

Subnet 4: 192.168.1.192/26

- What is the broadcast address of the third subnet?

Third subnet = 192.168.1.128/26

Broadcast address = 192.168.1.191

- How many usable hosts per subnet after splitting?

$2^{(32-26)} - 2 = 2^6 - 2 = 62$ usable hosts

Problem 2

You are a network admin for a small office. You have these departments:

- Sales: needs 30 hosts
- Engineering: needs 50 hosts
- HR: needs 10 hosts
- Servers: needs 10 hosts

You are given the network 10.0.0.0/24.

- Design a subnetting scheme that gives each department enough addresses.

Department	Network Address	CIDR	Usable Host Range	Broadcast Address
Engineering	10.0.0.0	/26	10.0.0.1 - 10.0.0.62	10.0.0.63
Sales	10.0.0.64	/27	10.0.0.65 - 10.0.0.94	10.0.0.95
HR	10.0.0.96	/28	10.0.0.97 - 10.0.0.110	10.0.0.111
Servers	10.0.0.112	/28	10.0.0.113 - 10.0.0.126	10.0.0.127

- For each department, write: network address, subnet mask (CIDR), usable host range, broadcast address.
- Which department gets the largest subnet? Why?
Engineering gets the largest subnet because it requires 50 hosts, which is more than any other department

Problem 3 — Connect to Security

- Why would a security team want to put the Servers department on a different subnet from Sales? (Answer using CIA + AAA concepts)

A security team would place the Servers department on a different subnet from Sales to improve security. Servers usually store important company data, so separating them helps protect confidentiality and integrity. Authorization rules can be applied so that only approved users and systems can access the servers. If a Sales computer is infected with malware, subnet separation also helps protect the availability of server resources.

- If an attacker compromises a Sales PC, can they directly access a Server on a different subnet? Explain why or why not.

If an attacker compromises a Sales PC, they cannot automatically access a server on a different subnet. Traffic between subnets is normally inspected by routers and firewalls. Security policies may block unauthorized access attempts. The attacker would need additional permissions or vulnerabilities to reach protected systems.

- What device controls traffic between subnets? (Hint: think about Day 2 — routers/firewalls)
The device that controls traffic between subnets is usually a router or a firewall. These devices examine network traffic and apply security rules. They help enforce access control and network segmentation.

Part 3 — Connect Everything: CIA + AAA Across All 4 Days

This is the thinking part. Write short paragraphs (3-5 sentences each). No copy-paste — use your own words.

Question 1 — CIA Triad in Action

Pick one of the protocols you studied (HTTP, FTP, SMTP, DNS, SSH, or ICMP). For that protocol:

- How does it support one part of the CIA Triad?
- How does it fail to protect one part of the CIA Triad?
- What would a security professional do to fix that weakness?

I chose HTTP as the protocol. HTTP supports availability because it allows users to access websites and web applications easily. However, HTTP does not provide confidentiality because data is sent in plain text and can be captured by attackers. A security professional would replace HTTP with HTTPS, which uses TLS encryption to protect data during transmission and improve confidentiality.

Question 2 — AAA Framework in Action

Think about logging into a website (like Gmail or any site you use).

- Authentication: How does the site verify you are who you say you are?
- Authorization: After you log in, what are you allowed to do? What are you NOT allowed to do?
- Accounting: What records might the site keep about your login? Why?

Authentication occurs when a website verifies a user through a username, password, and sometimes multi-factor authentication. This ensures that the person logging in is the legitimate account owner.

Authorization determines what actions a user can perform after logging in. For example, a Gmail user can access their own emails but cannot access another user's inbox. Permissions are granted based on the user's role and account privileges.

Accounting keeps records of login times, IP addresses, browser information, and user activities. These logs help security teams investigate incidents and monitor suspicious behavior.

Now answer: What happens if an attacker captures your HTTP traffic from Part 1? Which part of AAA breaks first? Why?

If an attacker captures HTTP traffic, Authentication is the first AAA component to fail. Because HTTP is unencrypted, usernames and passwords may be exposed. The attacker can then use the stolen credentials to impersonate the victim and gain unauthorized access.

Question 3 — The Full Attack Story

Imagine this scenario:

An employee at a company uses an unencrypted HTTP connection to log into the internal HR portal from a coffee shop. The company's network is poorly segmented — all departments are on the same subnet 192.168.1.0/24.

Using everything you learned this week, write a short attack story:

1. What can the attacker on the coffee shop Wi-Fi see? (Day 3 — HTTP, packet capture)

An employee connects to a coffee shop Wi-Fi network and accesses the company HR portal using an unencrypted HTTP connection. An attacker on the same Wi-Fi network captures packets using a packet analysis tool and is able to view the employee's username and password in plain text.

2. What can the attacker do with those credentials? (Day 1 — Authentication broken, CIA violated)

After obtaining the credentials, the attacker logs into the HR portal as the employee. This breaks authentication and violates confidentiality because sensitive company information is exposed to an unauthorized person.

3. Once inside the company network, what can the attacker reach? (Day 3 — subnetting, no segmentation)

The company network is poorly segmented and all departments are located on the same subnet. Once inside the network, the attacker can scan other devices, discover shared resources, and potentially access systems belonging to HR, Sales, Engineering, and other departments.

4. How could the company have prevented this? (Day 4 — hardening, Day 3 — segmentation, Day 1 — controls)

The company could have prevented this attack by using HTTPS instead of HTTP, enabling multi-factor authentication, applying strong access controls, and separating departments into different subnets protected by firewalls. These measures would reduce the attack surface and limit an attacker's ability to move through the network.

Question 4 — Linux/Windows Connection

- On Day 4 you learned about file permissions (Linux) and Event Viewer (Windows).
- How does the principle of least privilege (Day 4) connect to the AAA Authorization function (Day 1)?

The principle of least privilege means users should receive only the permissions required to perform their tasks. This directly supports the Authorization component of AAA because users are granted only the access they need and nothing more.

- If an attacker gets a shell on a Linux machine, what permission-related files would they look for?

If an attacker gains a shell on a Linux machine, they may examine files such as `/etc/passwd`, `/etc/shadow`, and `sudo` configuration files to identify accounts, permissions, and possible privilege escalation opportunities.

- On Windows, where would you look in Event Viewer to spot a brute-force login attempt?

On Windows, Event Viewer can be used to identify brute-force login attempts. Security analysts typically review Windows Logs → Security and look for repeated failed login events, authentication failures, and suspicious login activity.

How to Submit

Create a single document (text file, markdown, or PDF) with:

1. Part 1: Your traffic analysis table + answers + at least 2 screenshots from Wireshark
2. Part 2: All subnetting answers with working shown
3. Part 3: All 4 written answers in your own words

Name it: Week1_Weekend_Assignment_YourName.md (or .txt or .pdf)

Grading Rubric (self-grade before submitting)

Criteria	Points	Your score
Part 1: Traffic captured and analyzed correctly	/25	
Part 1: CIA + AAA connections made	/10	
Part 2: Subnetting calculations correct	/25	
Part 2: Security reasoning about subnets	/10	
Part 3: Own words, no copy-paste	/15	
Part 3: Connections across all 4 days are accurate	/15	
Total	/100	

Tips for Doing This Well

- Do Part 1 first. It is the most hands-on and will make the theory in Parts 2 and 3 feel real.
- Use your notes from all 4 days. This assignment is designed to make you go back and review everything.

- If you get stuck on subnetting, re-read the Day 3 notes. The 192.168.1.0/24 example is your starting point.
- Screenshots matter. In cybersecurity, evidence matters. Show what you captured.
- Write like you are explaining to a friend. If you cannot explain it simply, you do not understand it yet.

Week 1 — Assignment 2

"Build Your Home Lab & Security Audit"

Due: Alongside the Weekend Assignment | Time estimate: 2-3 hours

What you need: Your own laptop/PC, this document, internet

Why This Assignment Exists

The first assignment was about capturing and analyzing traffic. This one is about understanding your own environment — the machine you use every day. A security professional who does not understand their own system cannot protect anyone else's.

You will:

1. Audit your own computer's network configuration
2. Check what is running and listening on your machine
3. Analyze your own network topology
4. Write a mini security assessment of your home network

Task 1 — Your Machine's Network Identity (30 min)

Step 1: Gather your network information

On Windows: Open Command Prompt and run: `ipconfig /all`

On Linux/Mac: Open Terminal and run: `ifconfig` or `ip a`

Step 2: Record the following

Item	Your value
Your computer's hostname	DESKTOP-4B7CDOD
Your IPv4 address	10.61.36.71
Your subnet mask	255.255.255.0
Your default gateway (router IP)	10.61.36.179
Your DNS server(s)	10.61.36.179
Your MAC address	40-74-E0-C9-03-E7
Are you on Wi-Fi or Ethernet?	Wi-Fi

Step 3: Answer these questions

1. What is your network's IP range? (Hint: if you are 192.168.1.105/24, your range is 192.168.1.0 – 192.168.1.255)
10.61.36.71
2. How many devices could potentially be on your network?
254 usable devices.
3. Is your DNS server your router or an external service?
The DNS server appears to be the router/local network DNS service.
4. Why does the MAC address matter? (Hint: it is a Layer 2 identifier — visible even before IP communication)
The MAC address is a Layer 2 (Data Link Layer) identifier used for communication within the local network. Before IP communication occurs, devices use ARP to resolve an IP address into a MAC address. Every network adapter has a unique MAC address, making it important for device identification and local network communication.

Task 2 — What Is Listening on Your Machine? (30 min)

Step 1: Check open ports

On Windows: `netstat -an`

On Linux/Mac: `ss -tlnp`

Local Address	Port	State	Service
0.0.0.0	135	LISTENING	RPC Endpoint Mapper
0.0.0.0	445	LISTENING	SMB File Sharing
0.0.0.0	3306	LISTENING	MySQL Database
0.0.0.0	1521	LISTENING	Oracle Database
127.0.0.1	5500	LISTENING	VS Code Live Server
127.0.0.1	2030	LISTENING	Local Application Service

Step 2: Record what you find

Local Address	Port	State	What service do you think this is?
0.0.0.0	135	LISTENING	RPC Endpoint Mapper
0.0.0.0	3306	LISTENING	MySQL Database
127.0.0.1	5500	LISTENING	VS Code Live Server
127.0.0.1	2030	LISTENING	Local Application Service

Step 3: Answer these questions

1. How many ports are in LISTENING state?

More than 20 listening ports were observed on the system.

2. Are any of these ports from the Common Ports list in your cheat sheet?

Port	Service
135	RPC
139	NetBIOS
445	SMB
3306	MySQL
53	DNS
443	HTTPS (Established Connections)

3. Is port 3389 (RDP) open? If yes, what does that mean for security?

No Port 3389 was not found in the LISTENING state

4. Is port 23 (Telnet) open? If yes, why is that a problem?

No port 23 was not observed.

5. What is the difference between 0.0.0.0:PORT and 127.0.0.1:PORT?

0.0.0.0: Port:

Service listens on all network interfaces.

Other devices may be able to connect.

127.0.0.1:Port:

Localhost only.

Only your own computer can access it.

Step 4: Check running processes

On Windows: tasklist

On Linux/Mac: ps aux

List the top 10 processes you see. Identify any you do not recognize.

Process	Purpose
System Idle Process	CPU idle time
System	Windows Kernel
Secure System	Virtualization-based security
Registry	Windows Registry management
smss.exe	Session Manager

Process	Purpose
csrss.exe	Client/Server Runtime
wininit.exe	Windows initialization
services.exe	Service Control Manager
Lsass.exe	Credential Guard security
lsass.exe	Authentication and logon security

Task 3 — Scan Your Home Network (30 min)

Step 1: Find your network range

From Task 1, you know your IP and subnet mask. Calculate your network range.

Example: If your IP is 192.168.1.105 and subnet mask is 255.255.255.0:

- Network: 192.168.1.0/24
- Range: 192.168.1.1 to 192.168.1.254

IPv4 Address: 10.61.36.71

Subnet Mask: 255.255.255.0

Network: 10.61.36.0/24

Usable Host Range: 10.61.36.1 – 10.61.36.254

Step 2: Ping sweep your network

On Windows (PowerShell):

```
1..254 | ForEach-Object { ping -n 1 -w 100 192.168.1.$_ | Select-String "Reply from" }
```

On Linux/Mac (Terminal):

```
for i in {1..254}; do ping -c 1 -W 1 192.168.1.$i | grep "bytes from"; done
```

Note: Replace 192.168.1 with YOUR network prefix. This scan is safe — ping only checks if a host responds.

Step 3: Record what you find

IP Address	Responds to ping?	Device type (if you know)
10.61.36.179	No	Ping is blocked

Step 4: Answer these questions

1. How many devices responded to your ping sweep?

0 devices responded because the institutional network appears to block ICMP echo replies.

2. Did you find any devices you did not expect?

No unexpected devices were identified because the ping sweep did not receive responses.

3. If an attacker got onto your Wi-Fi, how many devices could they potentially reach?

The network range is 10.61.36.0/24, which supports up to 254 usable host addresses.

4. What is the security risk of having all devices on one flat network?

An attacker who compromises one device could potentially scan and communicate with many other devices on the same subnet. This increases the risk of lateral movement, unauthorized access, and malware propagation. Network segmentation reduces this risk

Task 4 — Your Home Network Security Assessment (30 min)

Write a mini security assessment of your own home network. Use everything you learned this week. Write in your own words. Minimum 300 words.

Structure your assessment like this:

1. Network Overview — What is your network range? How many devices? What types?
My system is connected to a Wi-Fi network with the IPv4 address 10.61.36.71 and subnet mask 255.255.255.0. This gives a network range of 10.61.36.0/24, supporting up to 254 usable hosts. The default gateway and DNS server are both configured as 10.61.36.179. The network contains devices such as laptops, mobile phones, and network infrastructure devices. A ping sweep was attempted, but no devices responded because ICMP echo replies appear to be blocked on the network.
2. Open Ports & Services — What ports are open? Which services are running?
A port scan using netstat revealed several listening ports including 135, 139, 445, 1521, 3306, and 33060. Port 135 is associated with Microsoft RPC services, while ports 139 and 445 are used for Windows file sharing. Port 3306 indicates that MySQL database services may be installed on the system. Port 1521 is commonly associated with Oracle database services. Some services were bound only to localhost (127.0.0.1), which is more secure because they cannot be accessed from external devices.
3. Protocol Security — What protocols does your network use? Any unencrypted protocols?
The network uses several protocols including TCP, UDP, DNS, HTTPS, and SMB. HTTPS provides encrypted communication and helps protect sensitive information. DNS is used for domain name resolution. Some protocols such as SMB and unencrypted services could present security risks if not properly configured. Secure protocols should always be preferred over unencrypted alternatives.
4. CIA Triad Analysis — Confidentiality, Integrity, Availability of your network
Confidentiality is protected through encrypted protocols such as HTTPS. Integrity is maintained by ensuring that data is not modified by unauthorized users during transmission or storage. Availability is achieved by maintaining stable network connectivity and keeping services operational. Any malware infection, unauthorized modification, or denial-of-service attack could negatively affect one or more parts of the CIA Triad.
5. AAA Framework Analysis — Authentication, Authorization, Accounting on your network
Authentication verifies user identity through usernames and passwords. Authorization determines which resources users can access after logging in. Accounting records user activities such as login times, network access, and system events. These three functions help maintain accountability and improve overall network security
6. Recommendations — List 5 specific improvements you could make

#	Improvement	Why it matters	How to do it
1	Use strong Wi-Fi passwords	Prevent unauthorized access	Enable WPA2/WPA3 with a strong password
2	Keep software updated	Fix security vulnerabilities	Install Windows and application updates regularly
3	Enable firewall protection	Block unwanted connections	Use Windows Defender Firewall
4	Disable unused services	Reduce attack surface	Remove or stop unnecessary services
5	Use antivirus protection	Detect malware threats	Keep Microsoft Defender enabled and updated

Conclusion

Overall, the network appears functional and reasonably secure. However, reducing unnecessary services, maintaining updates, and following security best practices can further improve confidentiality, integrity, and availability. Regular monitoring and security awareness are important for protecting systems from modern cyber threats.

Task 5 — Bonus: Protocol Journal (optional, +10 bonus points)

For one full day, keep a protocol journal. Every time you use the internet, write down:

Time	What I did	Protocol used	Port	Encrypted?	What would an attacker see?
09:00	Watched YouTube	HTTPS	443	Yes	Encrypted traffic only
10:00	Used ChatGPT	HTTPS	443	Yes	Only domain via DNS

10:30	Opened WhatsApp Web	HTTPS	443	Yes	Messages remain encrypted
11:00	Downloaded VS Code Extension	HTTPS	443	Yes	Download details encrypted
11:10	Checked Gmail	HTTPS	443	Yes	Only destination domain visible
11:45	Accessed College Portal	HTTPS	443	Yes	Login credentials protected
12:30	Hotstar streaming	HTTPS	443	Yes	Stream content encrypted
15:30	Watched YouTube	HTTPS	443	Yes	Encrypted traffic only
18:15	Used Chatgpt	HTTPS	443	Yes	Only domain via DNS
20:00	Checked mail	HTTPS	443	Yes	Messages remain encrypted
22:30	Opened Whatsapp	HTTPS	443	Yes	Messages remain encrypted

Fill in at least 10 entries. This teaches you to think like a security analyst in your daily life.

How to Submit

Create a document with:

1. Task 1: Your network information table + answers
2. Task 2: Your port scan table + answers + process list
3. Task 3: Your ping sweep results + answers
4. Task 4: Your written security assessment (300+ words)
5. Task 5 (bonus): Your protocol journal (10+ entries)

Name it: Week1_HomeLab_Audit_YourName.md (or .txt or .pdf)

Grading Rubric

Criteria	Points	Your score
Task 1: Network identity recorded correctly	/15	
Task 1: Subnetting questions answered	/10	

Criteria	Points	Your score
Task 2: Port scan completed and analyzed	/15	
Task 2: Process identification	/5	
Task 3: Ping sweep completed	/10	
Task 3: Security reasoning about flat networks	/10	
Task 4: Written assessment quality (300+ words)	/20	
Task 4: CIA + AAA connections accurate	/10	
Task 4: 5 practical recommendations	/5	
Bonus: Protocol journal (10+ entries)	+10	
Total	/100 (+10 bonus)	

Tips

- Do this assignment on your HOME network, not a public Wi-Fi.
- Take screenshots of your ipconfig, netstat, and ping sweep output.
- Be honest in your assessment. If your Wi-Fi has no password, write that down.
- The best security assessments are the ones that make you uncomfortable.